

Linear Regression

Model Definition

Define the model for a single-variable linear regression:

$$y_i = a + b \cdot x_i + \varepsilon_i$$

$$\hat{y}_i = a + b \cdot x_i$$

where:

- y_i is the actual output for the i -th observation.
- $\hat{y}_i$ is the predicted output for the i -th observation.
- ε_i is the error term for the i -th observation.

Objective Function: Sum of Squared Errors

The goal is to minimize the sum of squared errors f :

$$f = \sum_{i=1}^n (y_i - (a + b \cdot x_i))^2$$

or equivalently,

$$f(a, b) = \sum_{i=1}^n (y_i - (a + b \cdot x_i))^2$$

Minimization Condition

To minimize $f(a, b)$, take partial derivatives with respect to a and b and set them to zero:

$$\begin{cases} \frac{\partial f(a, b)}{\partial a} = 0 \\ \frac{\partial f(a, b)}{\partial b} = 0 \end{cases}$$

Applying the Partial Derivatives

1. Partial derivative with respect to a :

$$-2 \sum_{i=1}^n (y_i - a - b \cdot x_i) = 0$$

2. **Partial derivative with respect to b :**

$$-2 \sum_{i=1}^n x_i (y_i - a - b \cdot x_i) = 0$$

Setting Up the Equations

The following two equations are obtained:

$$\begin{cases} \sum_{i=1}^n y_i = a \cdot n + b \sum_{i=1}^n x_i \\ \sum_{i=1}^n y_i x_i = a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 \end{cases}$$

or, using summation notation:

$$\begin{cases} \sum y = a \cdot n + b \cdot \sum x \\ \sum yx = a \cdot \sum x + b \cdot \sum x^2 \end{cases}$$

Converting to Mean Form

Dividing through by n gives equations in terms of means:

$$\begin{cases} \bar{y} = a + b \cdot \bar{x} \\ \overline{yx} = a \cdot \bar{x} + b \cdot \overline{x^2} \end{cases}$$

Solving in Matrix Form

Express the equations in matrix notation:

$$\begin{pmatrix} 1 & \bar{x} \\ \bar{x} & \overline{x^2} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \bar{y} \\ \overline{yx} \end{pmatrix}$$

To solve for a and b :

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 & \bar{x} \\ \bar{x} & \overline{x^2} \end{pmatrix}^{-1} \begin{pmatrix} \bar{y} \\ \overline{yx} \end{pmatrix}$$

Solving Simple Linear Regression

$$y_i = a + b \cdot x_i + \varepsilon_i$$

$$\hat{y}_i = a + b \cdot x_i$$

$y_i \rightarrow$ actual output for i th observation.

$\hat{y}_i \rightarrow$ predicted output for i th observation.

$\varepsilon_i \rightarrow$ error term for i th observation.

$$f = \varepsilon_1^2 + \varepsilon_2^2 + \dots + \varepsilon_i^2$$

$$f = \sum_{i=1}^n (y_i - (a + b \cdot x_i))^2$$

$$f(a, b) = \sum_{i=1}^n [y_i - (a + b \cdot x_i)]^2$$

$$f(a, b) \rightarrow \min$$

$$\begin{cases} \frac{df(a,b)}{da} = 0 \\ \frac{df(a,b)}{db} = 0 \end{cases}$$

$$\begin{cases} -2 \sum_{i=1}^n (y_i - a - b \cdot x_i) = 0 \\ -2x_i \sum_{i=1}^n (y_i - a - b \cdot x_i) = 0 \end{cases}$$

$$\begin{cases} \sum_{i=1}^n y_i - \sum_{i=1}^n a - \sum_{i=1}^n x_i^2 = 0 \\ \sum_{i=1}^n y_i x_i - \sum_{i=1}^n a x_i - \sum_{i=1}^n b x_i^2 = 0 \end{cases}$$

$$\begin{cases} \sum y - a \cdot n - b \cdot \sum x = 0 \\ \sum yx - a \sum x - b \sum x^2 = 0 \end{cases}$$

$$\begin{cases} \bar{y} - a - b\bar{x} = 0 \\ \bar{y}\bar{x} - a\bar{x} - b\bar{x}^2 = 0 \end{cases}$$

$$\begin{cases} a + b\bar{x} = \bar{y} \\ a\bar{x} + b\bar{x}^2 = \bar{y}x \end{cases}$$

$$\begin{pmatrix} 1 & \bar{x} \\ \bar{x} & \bar{x}^2 \end{pmatrix} \times \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \bar{y} \\ \bar{y}x \end{pmatrix}$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 & \bar{x} \\ \bar{x} & \bar{x}^2 \end{pmatrix}^{-1} \times \begin{pmatrix} \bar{y} \\ \bar{y}x \end{pmatrix}$$

Rule for Matrix Multiplication

To multiply two matrices A and B , the number of **columns** in A must be equal to the number of **rows** in B . If:

- Matrix A has dimensions $m \times n$ (with m rows and n columns).
- Matrix B has dimensions $n \times p$ (with n rows and p columns).

Then the resulting matrix $C = A \cdot B$ will have dimensions $m \times p$.

Matrix Multiplication Process

For each element in the resulting matrix C , we calculate it as follows:

$$C[i, j] = \sum_{k=1}^n A[i, k] \cdot B[k, j]$$

This means each element $C[i, j]$ is the dot product of the i -th row of A with the j -th column of B .

Example 1: Simple Matrix Multiplication

Suppose we have two matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Both matrices are 2×2 , so they can be multiplied. The result $C = A \cdot B$ will also be a 2×2 matrix.

Let's compute each element in C :

1. **Calculate** $C[1, 1]$:

$$C[1, 1] = (1 \cdot 5) + (2 \cdot 7) = 5 + 14 = 19$$

2. **Calculate** $C[1, 2]$:

$$C[1, 2] = (1 \cdot 6) + (2 \cdot 8) = 6 + 16 = 22$$

3. **Calculate** $C[2, 1]$:

$$C[2, 1] = (3 \cdot 5) + (4 \cdot 7) = 15 + 28 = 43$$

4. **Calculate** $C[2, 2]$:

$$C[2, 2] = (3 \cdot 6) + (4 \cdot 8) = 18 + 32 = 50$$

So, the resulting matrix C is:

$$C = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Example 2: Multiplying Matrices with Different Dimensions

Suppose A is a 2×3 matrix, and B is a 3×2 matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$$

Here, A has 3 columns, and B has 3 rows, so we can multiply them. The result $C = A \cdot B$ will have dimensions 2×2 .

Let's calculate each element of C :

1. **Calculate** $C[1, 1]$:

$$C[1, 1] = (1 \cdot 7) + (2 \cdot 9) + (3 \cdot 11) = 7 + 18 + 33 = 58$$

2. **Calculate** $C[1, 2]$:

$$C[1, 2] = (1 \cdot 8) + (2 \cdot 10) + (3 \cdot 12) = 8 + 20 + 36 = 64$$

3. **Calculate** $C[2, 1]$:

$$C[2, 1] = (4 \cdot 7) + (5 \cdot 9) + (6 \cdot 11) = 28 + 45 + 66 = 139$$

4. **Calculate** $C[2, 2]$:

$$C[2, 2] = (4 \cdot 8) + (5 \cdot 10) + (6 \cdot 12) = 32 + 50 + 72 = 154$$

The resulting matrix C is:

$$C = \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix}$$

Summary of Rules

1. Matrices can only be multiplied if the number of columns in the first matrix equals the number of rows in the second matrix.
2. The resulting matrix will have dimensions that combine the rows of the first matrix with the columns of the second matrix. If A is $m \times n$ and B is $n \times p$, then $C = A \cdot B$ will be $m \times p$.
3. Each element $C[i, j]$ is calculated by taking the dot product of the i -th row of A with the j -th column of B .

Solving Multiple Linear Regression

To generalize your linear regression derivation for multiple independent variables (arguments), we extend the model to include k predictors, represented as $x_{i,1}, x_{i,2}, \dots, x_{i,k}$ for each observation i .

1. Model Definition:

The linear regression model with k predictors is:

$$y_i = a + b_1 \cdot x_{i,1} + b_2 \cdot x_{i,2} + \dots + b_k \cdot x_{i,k} + \varepsilon_i$$

Where:

- a : Intercept (bias term)
- $b_1, b_2, \dots, b_k$: Coefficients for predictors
- $x_{i,j}$: j -th predictor for the i -th observation
- ε_i : Error term

2. Predicted Output:

$$\hat{y}_i = a + \sum_{j=1}^k b_j \cdot x_{i,j}$$

3. Cost Function:

The residual sum of squares (RSS) is minimized:

$$f(a, b_1, b_2, \dots, b_k) = \sum_{i=1}^n \left[y_i - \left(a + \sum_{j=1}^k b_j \cdot x_{i,j} \right) \right]^2$$

4. Minimize Cost:

To find the parameters $a, b_1, b_2, \dots, b_k$, take partial derivatives with respect to a and each b_j , and set them to zero:

$$\frac{\partial f}{\partial a} = -2 \sum_{i=1}^n \left[y_i - \left(a + \sum_{j=1}^k b_j \cdot x_{i,j} \right) \right] = 0$$

$$\frac{\partial f}{\partial b_j} = -2 \sum_{i=1}^n x_{i,j} \cdot \left[y_i - \left(a + \sum_{j=1}^k b_j \cdot x_{i,j} \right) \right] = 0 \quad \text{for each } j = 1, 2, \dots, k$$

5. Solve the System of Equations:

Rearranging these equations leads to a linear system:

$$\begin{cases} \sum_{i=1}^n y_i = n \cdot a + \sum_{j=1}^k b_j \sum_{i=1}^n x_{i,j} \\ \sum_{i=1}^n y_i \cdot x_{i,j} = a \cdot \sum_{i=1}^n x_{i,j} + \sum_{m=1}^k b_m \sum_{i=1}^n x_{i,j} \cdot x_{i,m} \quad \forall j = 1, \dots, k \end{cases}$$

In matrix form:

$$\begin{pmatrix} n & \sum_{i=1}^n x_{i,1} & \sum_{i=1}^n x_{i,2} & \cdots & \sum_{i=1}^n x_{i,k} \\ \sum_{i=1}^n x_{i,1} & \sum_{i=1}^n x_{i,1}^2 & \sum_{i=1}^n x_{i,1}x_{i,2} & \cdots & \sum_{i=1}^n x_{i,1}x_{i,k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^n x_{i,k} & \sum_{i=1}^n x_{i,k}x_{i,1} & \sum_{i=1}^n x_{i,k}x_{i,2} & \cdots & \sum_{i=1}^n x_{i,k}^2 \end{pmatrix} \cdot \begin{pmatrix} a \\ b_1 \\ b_2 \\ \vdots \\ b_k \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n y_i \cdot x_{i,1} \\ \sum_{i=1}^n y_i \cdot x_{i,2} \\ \vdots \\ \sum_{i=1}^n y_i \cdot x_{i,k} \end{pmatrix}$$

6. Solution Using Matrix Notation:

Let:

- $\mathbf{X}$: $n \times (k + 1)$ design matrix (with 1 in the first column for the intercept)
- $\mathbf{y}$: n -dimensional column vector of outputs
- $\beta = (a \quad b_1 \quad \cdots \quad b_k)^T$: Coefficients

The normal equation is:

$$\mathbf{X}^T \mathbf{X} \cdot \beta = \mathbf{X}^T \mathbf{y}$$

Solve for β :

$$\beta = (\mathbf{X}^T \mathbf{X})^{-1} \cdot \mathbf{X}^T \mathbf{y}$$